

Supplementary Statistical Methods

Novel vs. older regimens for CRE infection in liver transplant recipients (CRE-COOLT)

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Study sample and endpoints

Analyses included the [r 405](#) liver transplant recipients with a documented CRE infection within 90 days of transplantation (one record per patient). The **primary endpoint** was all-cause mortality within 90 days; **secondary endpoints** were CRE infection recurrence and emergence of resistance to the administered regimen. Treatment was classified as *novel* (ceftazidime–avibactam, meropenem–vaborbactam, or cefiderocol) versus *older* (high-dose carbapenem-based) regimens.

Time origin, follow-up, and censoring

The time origin for all time-to-event analyses was the **date of CRE infection**. Event times were taken from the recorded dates of death, recurrence (relapse), and resistance emergence. Patients without an event were censored at their last valid contact date (the latest of hospital discharge, microbiological clearance, or end-of-therapy dates), and follow-up was administratively censored at 90 days.

Data-quality handling. Discharge dates recorded *before* the infection date ($n = 54$) were treated as invalid; for these records, censoring fell back to the latest remaining valid contact date. The competing-event (death) time was computed from the raw dates (`death_date - infection_date`) to provide a single, transparent source of timing.

Positivity and the era restriction

Novel agents were unavailable before 2018, so treatment assignment was perfectly separated by calendar era in the full cohort: no patient treated in 2010–2017 could have received a novel regimen. This is a structural violation of the **positivity (overlap) assumption**, under which a covariate that perfectly predicts treatment cannot be balanced by weighting. Accordingly, **all weighted analyses were restricted to the contemporaneous 2018–2025 era ($n = 182$)**, in which both regimens were used and outcome surveillance was concurrent. Full-cohort weighted estimates are not reported as primary because they conflate treatment with calendar era.

Propensity score and weighting

A propensity score (PS) for receipt of a novel regimen was estimated within the 2018–2025 cohort by logistic regression (`stats::glm`, binomial link) on four baseline covariates: **age, MELD at transplantation, multisite colonization, and SOFA score**. Stabilized inverse-probability-of-treatment weights (IPTW) targeting the average treatment effect were computed as $w = \hat{p}/\hat{e}$ for novel-treated patients and $w = (1 - \hat{p})/(1 - \hat{e})$ for older-regimen patients, where $\hat{e}$ is the estimated PS and $\hat{p}$ the marginal probability of novel treatment. Covariate balance was assessed by standardized mean differences (SMD), with $|\text{SMD}| < 0.10$ indicating adequate balance; after weighting, the maximum $|\text{SMD}|$ was 0.017 (Supplementary Figure 1).

Primary endpoint: 90-day mortality

The weighted treatment effect on 90-day mortality was estimated with a **weighted Cox proportional-hazards model** (`survival::coxph`, `weights = IPTW`, `robust = TRUE`), reported as a hazard ratio (HR) with robust 95% CI. The adjusted cumulative-mortality curve (Figure 1) is the complement of the weighted Kaplan–Meier estimate (`survival::survfit`), with 95% confidence bands from the robust variance. Because all-cause death is a terminal event with **no competing risk**, a subdistribution model is not applicable to mortality and was not used for this endpoint.

Subgroups. Consistency of the mortality effect was examined across pre-specified subgroups (carbapenemase status, infection source, SOFA score, therapy type) by fitting the weighted Cox model within each stratum and testing treatment-by-subgroup interactions (Supplementary Figure, forest plot). Subgroup sizes within the era are small; confidence intervals are correspondingly wide and interaction tests are of limited power.

Secondary endpoints: recurrence and resistance (competing risks)

Recurrence and resistance cannot occur after death, so death is a genuine **competing event**. Each patient was assigned a status of event of interest, death without the event, or censored. Weighted **Fine-Gray subdistribution hazard models** were fitted using `survival::finegray` to construct the subdistribution risk set, followed by `survival::coxph` with the product of the Fine-Gray censoring weights and the IPTW, and cluster-robust standard errors (`cluster(record_id)`). Effects are reported as subdistribution hazard ratios (sHR) with 95% CI. Cumulative incidence functions (Supplementary Figure 2) were estimated by weighted multi-state `survfit` (Aalen-Johansen), with death as the competing state.

Outcome ascertainment (interpretive caveat)

Recurrence and resistance require repeat cultures and susceptibility testing to be detected. In the 2010–2017 records these outcomes were rarely recorded (no resistance events; few recurrences; almost no microbiological clearance dates), consistent with less systematic retrospective follow-up. Crude, era-spanning rates of these endpoints are therefore susceptible to differential detection; the competing-risks comparison was based on the 2018–2025 era, in which surveillance was concurrent.

Software

All analyses were performed in **R version** `r as.character(getRversion())` (R Foundation for Statistical Computing). Analyses are fully reproducible from a single script (`analysis.R`).

Package	Version	Role
survival	3.8.9	Cox models, <code>finegray</code> , weighted <code>survfit</code> (competing risks, CIF)
WeightIt	1.7.0	Propensity-score weighting (verification)
cobalt	4.6.3	Covariate-balance diagnostics / Love plots
ggplot2	4.0.3	Figures
patchwork	1.3.2	Multi-panel figure composition
gt	1.3.0	Formatted tables
haven	2.5.5	Import of SPSS (<code>.sav</code>) data
tidyverse (dplyr)	2.0.0 (1.2.1)	Data management
scales	1.4.0	Axis formatting
svglite	2.2.2	Vector (SVG) figure export

All statistical tests were two-sided; $p < 0.05$ was considered statistically significant.